

Evan Jones
U.S. Olympic & Paralympic Committee Practicum
Bi-weekly progress report
Week ending March 9th

Mr. Schwartz,

In the past two weeks since the midterm progress report, I've made pretty significant strides in the visualization and user interface side of the project. I decided to take some time the first week to verify that the Tableau platform would be flexible enough to accommodate my planned visualizations. However, (as my slack message hinted at) it became quite clear that Tableau alone just wouldn't be sufficient to host my desired data visualizations. I researched a couple of alternative technologies and landed on using Dash. Dash is a python library that allows for interactive and dynamic plots in python all hosted via a local web server. After spending most of my week becoming familiar with the package, I was confident that this tool would be effective where Tableau fell short. After settling on continuing the visualization side of the project with Dash, I spent the following week working on building-out the web application itself. The application itself is now working nicely with a four-page layout with 3 unique and interactive dashboards (Page 1 is just a title and description page). So far, I have completely finished the title/landing page as well as the first dashboard used for visualizing an individual athlete's information, career, trajectory, etc. I also finalized some of the dashboard components used in the other 2 dashboards for country & event views. The most important of these components were my interactive KDE age plots which allow ad-hoc user hyperparameter tuning in application. In addition to these components, another significant task completed this week was the creation of all our necessary front end design elements. Although a bit of tedious work on the front-end, creating these style sheets and class designations with SCSS was an essential part of making the application have a strong and comfortable user interface. Additionally, I created a couple web scrapers to pull in athlete bio images and country flags which greatly add to the overall usability of the application. On the modeling side of things, I finished up my KDE analysis of Age from the perspective of medal achievement. However, inspired by your slack post about looking at age through the individual athlete lens, I've begun working on modeling from this perspective with the intent of comparing the results/difference/insights of my initial approach. I'm personally very interested to see if the different view of peak athlete performance changes the overall takeaways, and over the next two weeks I hope to narrow in on an answer to this question. Additionally, I'll also continue working on the next of my three dashboards, the event analysis dashboard. Thank for you continued help and support on this project. Let me know if you have any additional questions, and I will be happy to clarify.

Best regards,

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Evan Jones: United States Olympic & Paralympic Committee Revised Practicum Plan

Week	Week Ending	Planned	Completed	Other Notes
1	January 19 th	- Narrow sport scope to single discipline - Get familiarized with data structure and project goals - Exploratory data analysis with an eye towards data cleaning and formatting	- Alpine Skiing selected - Organized and learned about data structure - Identified droppable columns; Noticed fair amount of missing entries	
2	January 26 th	- Construct some histograms to test basic age to medal frequency and relationship - Research Alpine 12 unique events; sport governing bodies; scoring and ranking mechanics - Apply some cursory data cleaning and formatting - Identify and correct datatypes for database import	- Created histogram with clear relationship - Extensive notes and annotations taken on specific sport and event details - Dropped columns which were mostly null or near duplicates - Fixed pandas typing issues - Create coding system for missing, blank, nan, and other edge case field entries - Researched methods for filling missing data	
3	February 2 nd	- Create local MySQL database and load in data - Create python Selenium script for datamining International Ski Federation website for athletes - Separate and split team events from solo events - Compare histograms for different gender/event pairs to check similarities (see if sports should be combined or not)	- Local database created and working - Selenium script created and working; filled most missing values for age; scraped profile photos for visualization purposes later. - Data separated by team and gender groupings - Utility of combining event data still unclear further analysis needed	
4	February 9 th	- Organize data into format ready for KDE - Apply KDE to Men's Super G as a test run; Tune bandwidth; Test target variable approach medaling in general vs. specific medal. - Compile list of regulation changes for events - Add documentation to Selenium script/notebook	- Data formatted for KDE; initial results look promising. - Created bandwidth tuning interactive sliders for ad-hoc experimentation and visualization - Compiled list of impactful regulation changes and possible data adjustment remedies - Selenium documentation added	

5	February 16 th	- Research masking methods for train/test given problem setup - Apply remedies for regulations and necessary data adjustments - Fit KDE models on all solo events - Begin work on midterm slide deck	- Train & Testing years/masked determined - Decided to utilize only post reg data, but to include a dashboard view for better understanding of the impact of regulation changes - Fit KDE models on solo events - Developed diagrams for Midterm slide deck -	
6	February 23 rd	- Create country historic and forward-looking pipeline profiles which at each Olympic date contain competitors used; unused; and medal outcomes - Create selenium script for updating country pipelines as new athletes are entered in the FIS database. - Finish midterm slide deck	- Pipeline creation in progress, but not complete as of weekend - Finished midterm report presentation slides - Created flexible support for new athlete entries	Midterm slide deck/report due
7	March 2 nd	- Start layout and design work on Dash dashboard structure - Setup multi-page web app layout - Create landing page - Convert KDE models into dash friendly representation	- Multi-page dash app created - Finalized file structure for the application - Added global scss components and style sheets - Finalized application landing page - KDE models converted to dash interactive plot for customized hyperparameter selection.	Revised Project plan to utilize Dash instead of Tableau causes some changes in overall plan
8	March 9 th	- Create FIS image scraper for athlete profile images - Create all dash components and callbacks for the individual athlete view dashboard page - Create repo for country flags - Finish work on KDE descriptive stats and chi squared test	- Created image scraper used as design elements in dash - Created svg repository for displaying specific country and Olympic flags used as design elements in dash - Finished all callbacks and components on the individual athlete view dashboard - Began work on Event level dashboard - Took Elliot's advice about viewing age from the individual achievement perspective did some cursory modeling here - Finished KDE analysis	
9	March 16 th	- Finalize individual athlete peak achievement view of age analysis - Add visualization that contrasts the two different age modeling approaches.		

		- Start work on the event view dashboard		
10	March 23 rd	- Finish work on the event view dashboard - Begin country pipeline level analysis and prediction - Begin work on the last country view dashboard		
11	March 30 th	- Aggregate all production quality models and predictions - Detailed hyperparameter tuning - Pickle models for reuse - Finish country pipeline analysis - Finish country pipeline dash page		
12	April 6 th	- Cushion week for anything that went longer than expected		
13	April 13 th	- Organize and add documentation to repository - Cushion week for anything that went longer than expected		
14	April 20 th	- Prepare final report and touch up		Final report due